# Akoizo Original Practice - Form 5 - Answer key

### Division C | Mission Possible | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

### ANSWER KEY - Review only after submitting.

**1. (2 points) An IR action has the transmitter 15 cm from the receiver. Does it qualify?**

A. No; they must be at least 20 cm apart

B. Yes; any distance works

C. Yes, if the LED is visible light

D. No; they must be 50 cm apart

**Correct: A**

**2. (2 points) Which lever class has the load between the fulcrum and the effort?**

A. Second class

B. First class

C. Third class

D. None

**Correct: A**

**3. (2 points) What is the first tiebreaker?**

Unit: Where the points are: building a scoring plan.
Worked-case data: A Regional device starts on the first coin, completes 8 scorable actions and the Final Action at 58 s, earns all ASL points, sets up in 20 minutes, has no adjustments, and measures 60 x 60 x 70 cm. Estimate its score.
Separate investigation observations:
Plan A: 12 actions, works in 6 of 10 test runs.
Plan B: 9 actions, works in 10 of 10 test runs.
Failure cost: A failed run usually loses the Final Action, 75 no-adjust bonus, and some time points.

A. Fewest penalty points

B. Smallest device

C. Fastest setup

D. Most scorable actions

**Correct: A**

**4. (2 points) What must a State team do to earn water timer bonus points?**

Unit: Timing: Target Time, the water timer, and pacing.
Worked-case data: Target Time 60 s. The device runs 67 s with a 25-s water timer that ended at 40 s. Calculate time points and timer bonus.
Separate investigation observations:
Timer calibration: 150 mL -> 18 s; 300 mL -> 30 s; 450 mL -> 39 s.
Other actions: Total 46 s without the timer.
Target Time: 80 s announced at State.

A. Run the timer past the Target Time

B. Use an electric pump

C. Demonstrate how the timer is adjusted for the Target Time

D. Use a scorable action as the timer

**Correct: C**

**5. (4 points) How many moles of CO_2 are produced by 4.2 g of baking soda with excess vinegar?**

answer: 4.2 g ÷ 84 g/mol = 0.050 mol of CO_2, about 1.2 L at room temperature. (4 points).

**6. (4 points) Why must rejected coins be caught inside the device?**

Unit: The Start Action: coins, slots, and selective triggers.
Worked-case data: Design a rail gap that rejects nickels and dimes but passes quarters to a trigger, allowing 0.5 mm tolerance each side.
Separate investigation observations:
Gap 21.5 mm: Nickel sometimes stuck on the rail and triggered the start.
Gap 22.8 mm: Dime and nickel fell through 20 of 20; quarter carried 20 of 20.
Gap 24.0 mm: Quarter fell through in 6 of 20 trials.

answer: All coins must be returned to the supervisor, and a coin leaving the device would also cost the 50-point penalty. (4 points).

**7. (4 points) Why must the gum be removed after it lands?**

Unit: The Final Action, the ASL, and competition day.
Worked-case data: Estimate the landing speed of a gum pack dropped 55 cm, and suggest how to keep it on the platform.
Separate investigation observations:
Drop tests: Gum fell 52 cm; landed on platform 7 of 10 times, bouncing off 3.
Change: Added a level release shelf; dropped 51 cm.
New tests: Landed and stayed 10 of 10 times.

answer: The rules award the Final Action only if a team member removes the gum through the opening on the coin-slot side. (4 points).

**8. (4 points) Explain why lower resistance nichrome cuts line faster with the same battery.**

answer: Power is V^2/R, so lower resistance draws more current and dissipates more heat. (4 points).

**9. (4 points) Why does a first-class lever not always give a mechanical advantage greater than 1?**

answer: Its IMA depends on arm lengths; if the effort arm is shorter than the load arm, IMA is less than 1. (4 points).

**10. (6 points) Solve this case and explain your reasoning: A Regional device starts on the first coin, completes 8 scorable actions and the Final Action at 58 s, earns all ASL points, sets up in 20 minutes, has no adjustments, and measures 60 x 60 x 70 cm. Estimate its score.**

reasoning: Start 25 + actions 8 x 50 = 400 + Final 250 = 675.
Time 58 x 2 = 116; ASL 100; setup 50; no adjustments 75.
Size: 20 cm, 20 cm, and 10 cm under 80 -> 20 + 20 + 10 = 50. (3 points).

conclusion: Total about 675 + 116 + 100 + 50 + 75 + 50 = 1,066 points. (3 points).

**11. (6 points) Solve this case and explain your reasoning: Target Time 60 s. The device runs 67 s with a 25-s water timer that ended at 40 s. Calculate time points and timer bonus.**

reasoning: Time points: 2 x 60 = 120, minus 2 x 7 = 14 -> 106.
Water timer: ran 25 full seconds before the Target Time -> +25.
Total from timing: 131 points. (3 points).

conclusion: Shortening the timer by 7 s would raise time points to 120 while losing 7 timer points: a net gain of 7. (3 points).

**12. (2 points) What explains the results?**

Supplied synthetic observations:
Test 1: 1.0 g baking soda + 30 mL vinegar: balloon expanded 14 cm.
Test 2: 2.0 g + 30 mL vinegar: expanded 21 cm.
Test 3: 3.0 g + 30 mL vinegar: expanded 22 cm; undissolved powder left over.

A. Vinegar became the limiting reactant above about 2 g of baking soda

B. More baking soda always gives more gas

C. The balloon was too large

**Correct: A**

**13. (2 points) Define “Coin Drop Order” in the context of Mission Possible.**

answer: Nickel then dime, or dime then nickel, announced at setup. (2 points).

**14. (2 points) Define “Final Action” in the context of Mission Possible.**

answer: Gum pack dropped at least 50 cm onto the platform; 250 points. (2 points).

**15. (2 points) Define “Photodiode” in the context of Mission Possible.**

answer: Light sensor that produces current when light hits it. (2 points).

**16. (2 points) Define “Sodium bicarbonate” in the context of Mission Possible.**

answer: Baking soda; releases CO_2 with acids. (2 points).

**17. (2 points) Define “Supporting segments” in the context of Mission Possible.**

answer: Rope sections holding a pulley load; equal to IMA. (2 points).

**18. (2 points) Define “Action Sequence List (ASL)” in the context of Mission Possible.**

answer: Numbered documentation of every action, worth up to 100 points. (2 points).